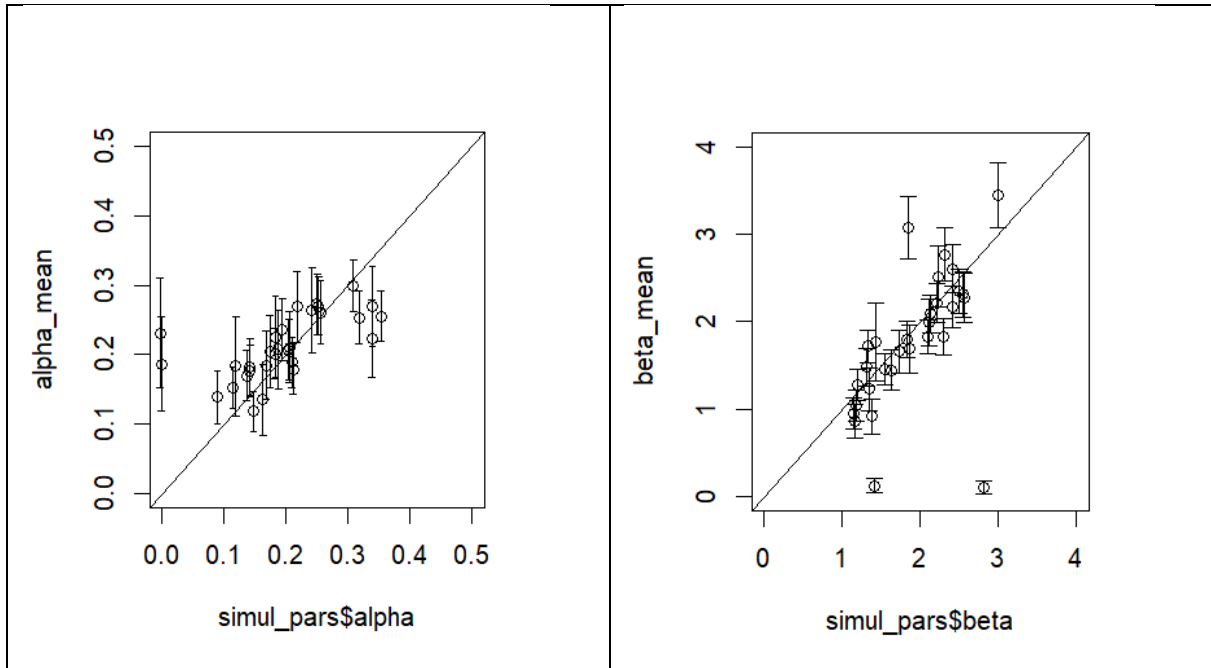


#1. (1) done with original R and Stan deleting some index(/)

(2a) done.

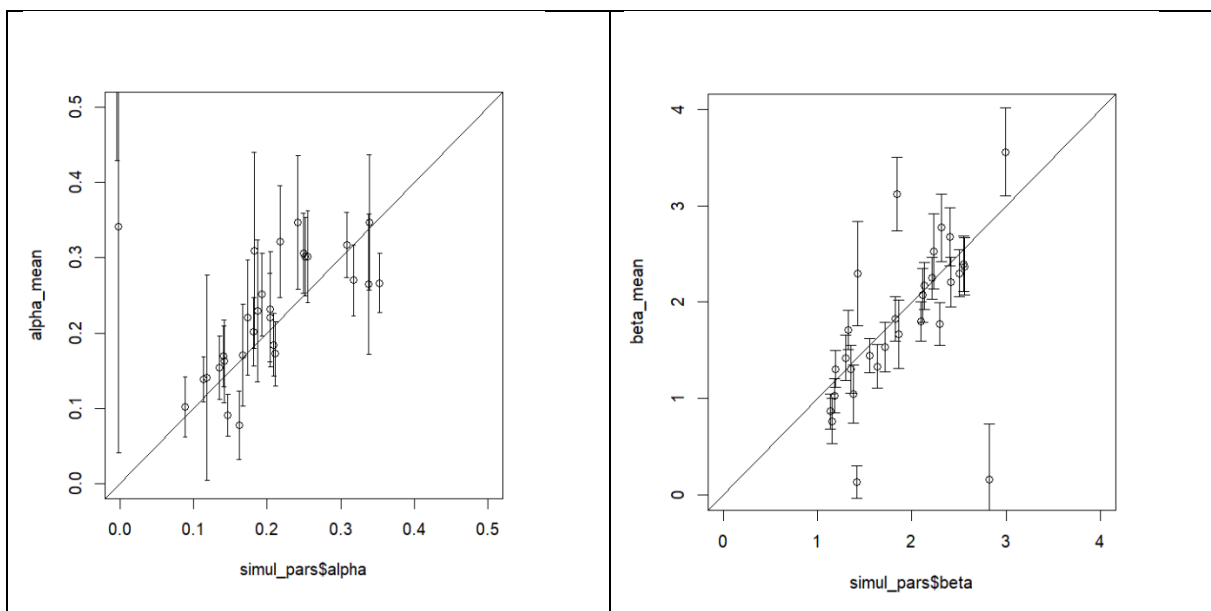
(2b)



(3)

#2. (1) hw6_model1_nohier and hw6_model1_nohier.stan are the files for this part

(2)

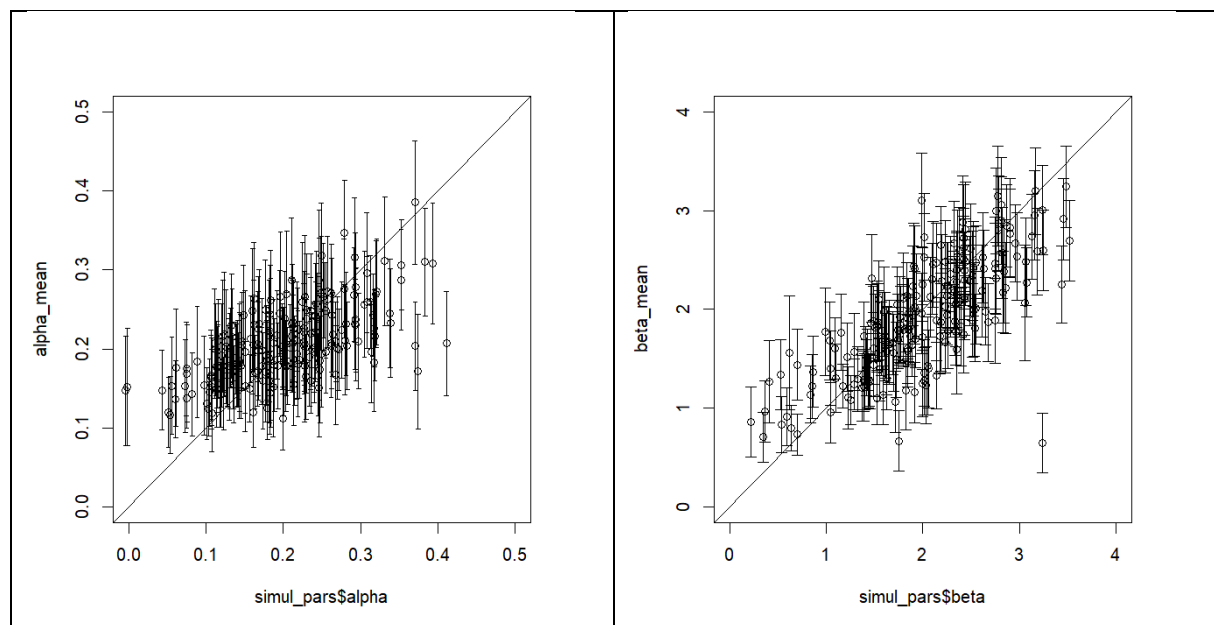


(3) The scatter plot from Question#1(2) used hierarchical model, which assumes that individual parameters are generated from group-level distribution. This makes the estimates of outliers shrink toward the group mean. However, non-hierarchical model(Question #2(2)) doesn't have shrinkage because they assumes individual parameters just as individuals, so the error bar is shorter in Question#1(2), which means that it is closer to the true parameter. There are some points which are totally far from the group mean in non-hierarchical model's scatter plots, which are not existing in hierarchical model's scatter plots.

#3. (1) `simulate_hw6_model1_fake` is used for this part

(2) done.

(3)

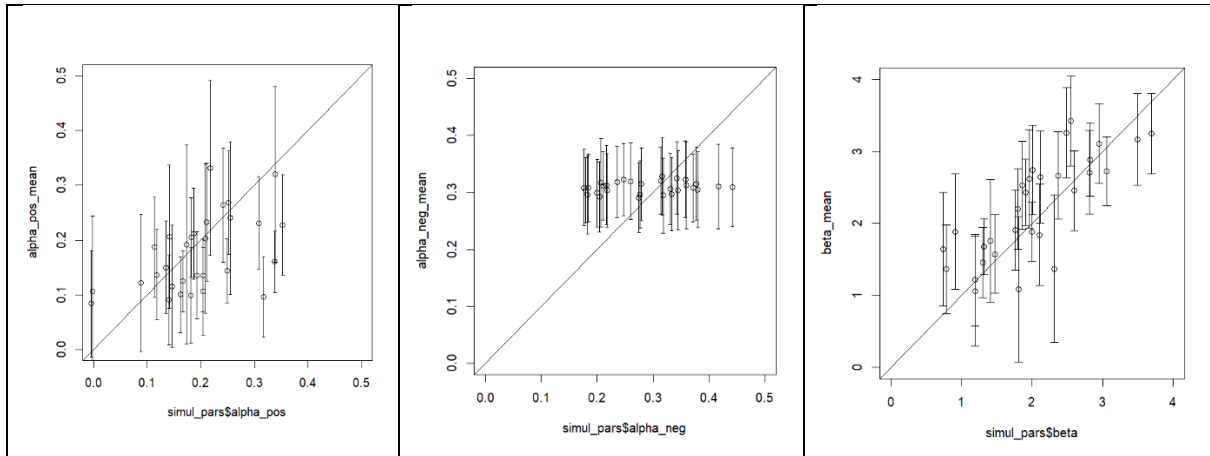


(4) Shrinkage happens when the likelihood provides weak information. Because not all the parameters explain the model equally, the parameter which gets lowest signal or high variance prone to shrink more, so beta showed the shrinkage while alpha, which provides stronger linkelihood information, got less shrinkage(I don't exactly know, but alpha also seems to have longer error bar, which may mean lower shrinkage)

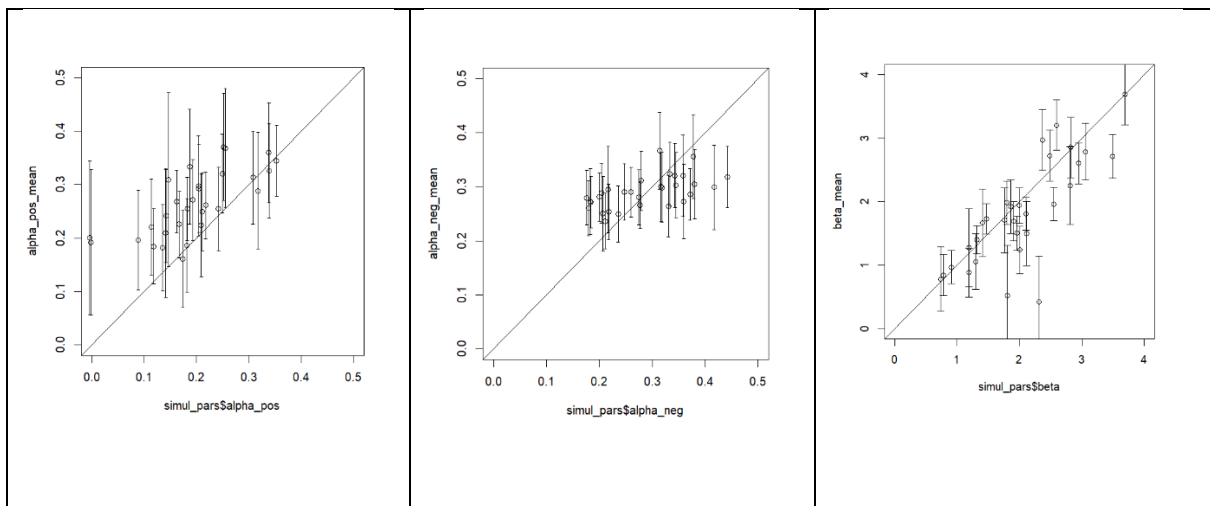
#4.

(1) the R and stan with `_rein` are used in this part

(2)



(3)



(4) In the result of question#4(2), Alpha positive and beta results showed shrinkage, but low trial numbers of 100 inhibited the shrinkage. However, In the result of question#4(3), both of them showed lower error bars and less outliers due to higher trial numbers of 300, which means more shrinkage. Also, alpha negative failed to recover in question#4(2), but it showed a little recovery in question#4(3) due to decreased positive probability from 0.5 to 0.3 and 0.8 to 0.3.